Project: Worked examples
Title: Motor and actuator sizing
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1. Scope and method

Torque, speed and power a motor needs to move a load through a mechanism on a point-to-point cycle: the load reflected to the motor shaft, the acceleration from the motion profile, peak torque during acceleration, RMS torque over the cycle (thermal duty), peak speed, inertia ratio, and a check of a candidate motor's rated and peak torque and speed with a safety margin. A rotary load that runs continuously (a pump, a fan) is sized instead from its steady torque and speed and a run-up time.

WHAT IT CALCULATES	RELATION
Motor-to-load ratio of a linear mechanism, in metres per motor radian	$k = p/(2\pi\, i)$ (screw); $D/(2\, i)$ (belt); $D_p/(2\, i)$ (rack)
Load inertia reflected to the motor through a linear mechanism	$J_{refl} = m\, k^2/\eta$
Load torque at the motor through a linear mechanism	$T_{load} = F\, k/\eta$, with $F = m\, g\, (\sin\theta + \mu \cos\theta) + F_p$
Reflected inertia and torque of a rotary load through a gearbox	$J_{refl} = J_L/(i^2\, \eta)$; $T_{load} = T_L/(i\, \eta)$
Move profile, trapezoidal 1/3–1/3–1/3 (adjustable)	$v_{pk} = s/(t_{move} - t_{acc})$; $a = v_{pk}/t_{acc}$
Move profile, S-curve (sine acceleration, same peak speed)	$a_{pk} = (\pi/2)\, a_{trap}$
Torque to accelerate the total inertia, and the peak torque	$T_{acc} = J_{total}\, \alpha$; $T_{pk} = T_{acc} + T_{load}$
Alternating torque amplitude of a point-to-point cycle about T_{load}, for a shaft's fatigue	$T_{alt} = (T_{pk} - T_{dec})/2$ (0 in continuous duty). Torque behind the ratio on the output shaft of a rotary mechanism: $T_{out} = T \cdot i \cdot \eta$ (mean, alternating and peak), $n_{out} = n_{pk}/i$. Peak axial thrust of a ball screw on its support bearings: $F_{ax,pk} = F_{load} + m\, a_{pk}$ (0 for other mechanisms)
RMS (thermal) torque over the whole cycle, dwell included	$T_{rms} = \sqrt{(\Sigma\, T_i^2\, t_i / t_{cycle})}$
RMS mechanical power over the cycle, the thermal-equivalent continuous power a supply sees (regeneration counts as power; the instantaneous peak P_{pk} is not a continuous demand), trapezoidal profile	$P_{rms} = \omega_{pk}\, \sqrt{\{[t_{acc} \cdot (2/3)(T_{load}^2 + T_{acc}^2) + T_{load}^2\, t_{const}]/t_{cycle}\}}$; for the S-curve the acceleration term becomes $t_{acc} \cdot [(3/4)\, T_{load}^2 + (5/16)\, T_{acc}^2]$
Continuous rotary duty (a pump or fan), speed at the motor	$\omega = 2\pi\, n_L\, i/60$; $n_{pk} = n_L\, i$
Continuous rotary duty, load torque and RMS torque	$T_{load} = T_L/(i\, \eta)$; $T_{rms} = T_{load}$
Peak torque during the run-up, with full load throughout (conservative)	$T_{pk} = J_{total}\, \omega/t_{start} + T_{load}$

WHAT IT CALCULATES	RELATION
Continuous power	$P_{cont} = T_{load}\,\omega$; $P_{rms} = P_{cont}$
Rated shaft power of the candidate motor, from its rated torque and its rated speed n_{rated} (not the running speed), the shaft power its supply must be sized for; without n_{rated} there is no P_{rated} and the sheet warns	$P_{rated} = T_{rated} \cdot 2\pi\, n_{rated}/60$

Basis: Load torque, inertia referred to the motor, duty cycles[1](Ch. on motor and drive selection); Reflected inertia, RMS torque, inertia ratio practice[2].

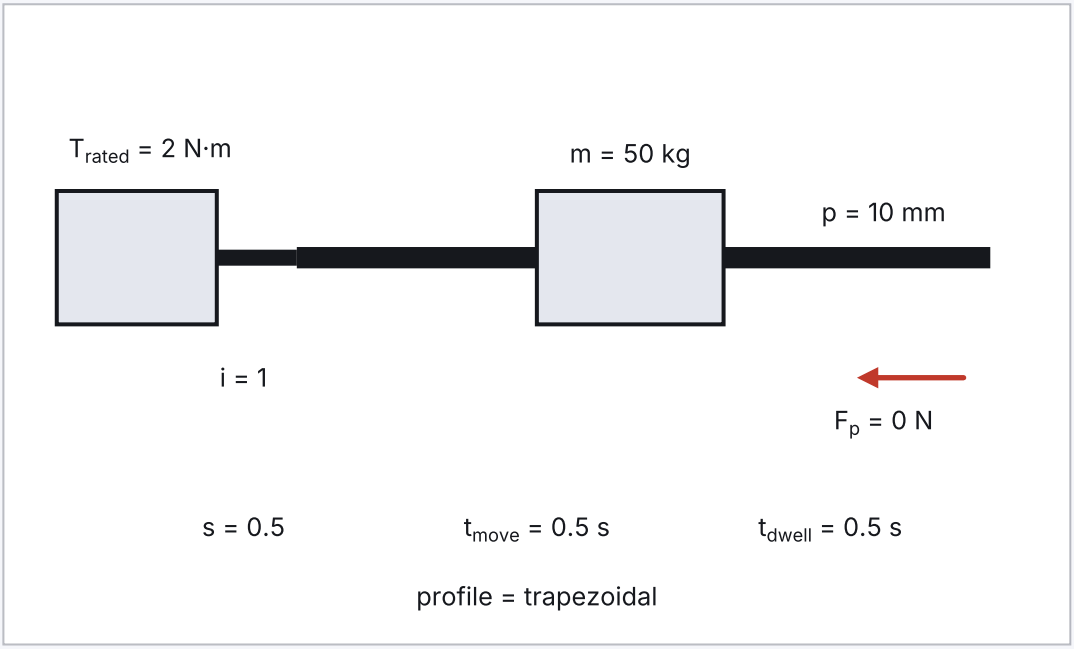


Figure 1 — A motor driving a lead or ball screw; the carriage of mass m moves along the guide. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mechanism	Mechanism	`lead / ball screw`		assumption: Sizing convention.	any
i	Gear ratio i (motor : output, 1 = none)	`1`		assumption: Sizing convention.	1 … 1000
η	Mechanism efficiency (overall)	`0.9`		spec: Screw datasheet BS-1610: 90 %	0.3 … 1
p_{screw}	Screw lead (per revolution of the screw)	`10`	mm	spec: Screw datasheet BS-1610: 10 mm lead	1 … 100 mm
J_{mech}	Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)	`0.0001`	kg·m²	spec: Screw datasheet BS-1610 + coupling: 1.0e-4 kg·m²	0 … 10 kg·m²

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
m	Moved mass (linear mechanisms)	50	kg	spec: Mechanical layout ML-12: carriage + gripper 50 kg	0 … 100000 kg
μ	Friction coefficient (guide / bearing)	0.1		assumption: Linear guides with seals: μ = 0.1.	0 … 1
θ	Incline from horizontal (+ moving uphill)	0	°	assumption: Sizing convention.	-90 … 90 °
F_p	Process force opposing the move (linear)	0	N	assumption: Sizing convention.	0 … 1000000 N
T_L	Constant load torque at the output (rotary)	0	N·m	default	0 … 100000 N·m
$duty_{mode}$	Duty (point-to-point move cycle, or continuous rotary running with a run-up)	point-to-point		default	any
profile	Motion profile	trapezoidal		assumption: Sizing convention.	any
s	Move distance (linear: m; rotary: output revolutions)	0.5		spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.0001 … 10000
t_{move}	Move time	0.5	s	spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.01 … 3600 s
f_{acc}	Acceleration (and deceleration) fraction of the move time	0.333333		spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.05 … 0.5
t_{dwell}	Dwell between moves	0.5	s	spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0 … 3600 s
J_{motor}	Candidate motor rotor inertia	0.0002	kg·m²	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	1e-7 … 10 kg·m²
T_{rated}	Candidate motor continuous (rated) torque	2	N·m	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	0.01 … 100000 N·m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
T_{peak}	Candidate motor peak torque	6	N·m	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	0.01 … 100000 N·m
n_{max}	Candidate motor maximum speed	10000	rpm	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	10 … 100000 rpm
SF	Safety margin on torque	1.25		assumption: Sizing convention.	1 … 3
$J_{ratio,max}$	Inertia ratio limit (load : motor), warning	10		assumption: Sizing convention.	1 … 100

3. Assumptions

A1. Rigid transmission; the mechanism efficiency applies to the driving direction and is also used in the deceleration phase (conservative for peak, slightly pessimistic for RMS). (Servo-sizing practice.)
A2. The friction force is Coulomb friction on the moved mass; viscous and breakaway friction are not modelled unless included in F_p. (Load model.)
A3. Continuous duty: the load torque is constant at the stated speed and the motor accelerates the whole inertia linearly to speed in t_{start} while carrying the full load torque (conservative); the move distance, move time, acceleration fraction, dwell and profile are ignored. (Run-up idealisation for pumps and fans.)
A4. The S-curve profile has sine-shaped acceleration over the same acceleration time, so its peak acceleration is π/2 times the trapezoidal value. (Jerk-limited profile idealisation.)
A5. mechanism (Mechanism) is taken as lead / ball screw. (Sizing convention.)
A6. i (Gear ratio i (motor : output, 1 = none)) is taken as 1. (Sizing convention.)
A7. μ (Friction coefficient (guide / bearing)) is taken as 0.1. (Linear guides with seals: μ = 0.1.)
A8. θ (Incline from horizontal (+ moving uphill)) is taken as 0 deg. (Sizing convention.)
A9. F_p (Process force opposing the move (linear)) is taken as 0 N. (Sizing convention.)
A10. profile (Motion profile) is taken as trapezoidal. (Sizing convention.)
A11. SF (Safety margin on torque) is taken as 1.25. (Sizing convention.)
A12. $J_{ratio,max}$ (Inertia ratio limit (load : motor), warning) is taken as 10. (Sizing convention.)

4. Working

[22] $k = p_{screw} / (2 * pi * i)$
$= 10\ mm / (2 \times \pi \times 1)$
$= 1.5915\ mm$
Travel per motor radian

[23] $F_{load} = m * 9.80665\ m/s^2 * (\sin(\theta) + \mu * \cos(\theta)) + F_p$
$= 50\ kg \times 9.80665\ m/s^2 \times (\sin(0\ deg) + 0.1 \times \cos(0\ deg)) + 0\ N$
$= 49.033\ N$
Load force along the motion (gravity, friction, process)

[24] $T_{load} = F_{load} * k / \eta$
$= 49.033\ N \times 1.5915\ mm / 0.9$
$= 0.08671\ N·m$
Load torque at the motor shaft

[25] $J_{refl} = m * k^2 / \eta$
$= 50\ kg \times (1.5915\ mm)^2 / 0.9$
$= 0.00014072\ kg·m²$
Load inertia reflected to the motor shaft

[26] $v_{pk} = s * 1\ m / (t_{move} * (1 - f_{acc}))$
$= 0.5 \times 1\ m / (0.5\ s \times (1 - 0.33333))$
$= 1.5\ m/s$
Peak linear velocity (trapezoid)
[27] $a_{trap} = v_{pk} / (f_{acc} * t_{move})$
$= 1.5\ m/s / (0.33333 \times 0.5\ s)$
$= 9\ m/s^2$
Trapezoidal acceleration
[28] $\omega_{pk} = v_{pk} / k$
$= 1.5\ m/s / 1.5915\ mm$
$= 942.48\ rad/s$
Peak motor speed
[30] $\alpha_{pk} = a_{trap}$
$= 9\ m/s^2$
$= 9\ m/s^2$
Peak acceleration
[31] $\alpha_{pk} = a_{pk} / k$
$= 9\ m/s^2 / 1.5915\ mm$
$= 5654.9\ rad/s^2$
Peak motor angular acceleration
[32] $J_{total} = J_{motor} + J_{mech} + J_{refl}$
$= 0.0002\ kg \cdot m^2 + 0.0001\ kg \cdot m^2 + 0.00014072\ kg \cdot m^2$
$= 0.00044072\ kg \cdot m^2$
Total inertia at the motor shaft
[33] $T_{acc} = J_{total} * \alpha_{pk}$
$= 0.00044072\ kg \cdot m^2 \times 5654.9\ rad/s^2$
$= 2.4922\ N \cdot m$
Acceleration torque
[34] $T_{pk} = T_{acc} + T_{load}$
$= 2.4922\ N \cdot m + 0.08671\ N \cdot m$
$= 2.5789\ N \cdot m$
Peak torque (accelerating against the load)
[35] $T_{dec} = -T_{acc} + T_{load}$
$= -2.4922\ N \cdot m + 0.08671\ N \cdot m$
$= -2.4055\ N \cdot m$
Torque during deceleration
[36] $T_{alt} = (T_{pk} - T_{dec}) / 2$
$= (2.5789\ N \cdot m - (-2.4055\ N \cdot m)) / 2$
$= 2.4922\ N \cdot m$
Alternating torque amplitude of the cycle about T_{load}
[37] $F_{ax,pk} = F_{load} + m * a_{pk}$
$= 49.033\ N + 50\ kg \times 9\ m/s^2$
$= 499.03\ N$
Peak axial thrust on the screw's support bearings (load force plus the moved mass accelerated)
[38] $n_{pk} = \omega_{pk}$
$= 942.48\ rad/s$
$= 9000\ rpm$
Peak motor speed
[39] $P_{pk} = T_{pk} * \omega_{pk}$
$= 2.5789\ N \cdot m \times 942.48\ rad/s$
$= 2.4306\ kW$
Peak mechanical power

[40] $t_{acc} = f_{acc} * t_{move}$
= 0.33333 × 0.5 s
= 0.16667 s
Acceleration time
[41] $t_{const} = t_{move} - 2 * t_{acc}$
= 0.5 s − 2 × 0.16667 s
= 0.16667 s
Constant-velocity time
[42] $t_{cycle} = t_{move} + t_{dwell}$
= 0.5 s + 0.5 s
= 1 s
Cycle time
[43] $P_{rms} = \omega_{pk} * \mathrm{sqrt}((t_{acc} * (2 / 3) * (T_{load}^2 + T_{acc}^2) + T_{load}^2 * t_{const}) / t_{cycle})$
= 942.48 rad/s × sqrt((0.16667 s × 2 / 3 × ((0.08671 N·m)^2 + (2.4922 N·m)^2) + (0.08671 N·m)
0.16667 s) / 1 s)
= 0.78414 kW
RMS mechanical power over the cycle (thermal-equivalent continuous power)
[44] $T_{rms} = \mathrm{sqrt}((T_{pk}^2 * t_{acc} + T_{load}^2 * t_{const} + T_{dec}^2 * t_{acc}) / t_{cycle})$
= sqrt(((2.5789 N·m)^2 × 0.16667 s + (0.08671 N·m)^2 × 0.16667 s + (-2.4055 N·m)^2 × 0.16667
= 1.4402 N·m
RMS torque over the cycle (thermal)
[45] $J_{ratio} = (J_{refl} + J_{mech}) / J_{motor}$
= (0.00014072 kg·m² + 0.0001 kg·m²) / 0.0002 kg·m²
= 1.2036
Inertia ratio (load : motor)
[46] $u_{peak} = SF * T_{pk} / T_{peak}$
= 1.25 × 2.5789 N·m / 6 N·m
= 0.53728
Peak torque utilisation with safety margin
[47] $u_{rms} = SF * T_{rms} / T_{rated}$
= 1.25 × 1.4402 N·m / 2 N·m
= 0.90012
Continuous torque utilisation with safety margin
[48] $u_{speed} = n_{pk} / n_{max}$
= 9000 rpm / 10000 rpm
= 0.9
Speed utilisation
[49] $duty = T_{rms} / T_{rated}$
= 1.4402 N·m / 2 N·m
= 0.7201
Thermal duty (without margin)

5. Results

SYMBOL	VALUE	DESCRIPTION
k	1.5915 mm	Transmission constant
T_{load}	0.08671 N·m	Load torque at motor
J_{refl}	0.00014072 kg·m²	Reflected load inertia
J_{total}	0.00044072 kg·m²	Total inertia
v_{pk}	1.5 m/s	Peak velocity
a_{pk}	9 m/s²	Peak acceleration
α_{pk}	5654.9 rad/s²	Motor angular acceleration

SYMBOL	VALUE	DESCRIPTION
T_{acc}	2.4922 N·m	Acceleration torque
T_{pk}	2.5789 N·m	Peak torque
T_{dec}	-2.4055 N·m	Deceleration torque
T_{alt}	2.4922 N·m	Alternating torque amplitude
$F_{ax,pk}$	499.03 N	Peak axial thrust on a screw's bearings
T_{rms}	1.4402 N·m	RMS torque
n_{pk}	9000 rpm	Peak speed
P_{pk}	2.4306 kW	Peak power
P_{rms}	0.78414 kW	RMS power (continuous equivalent)
J_{ratio}	1.2036	Inertia ratio
duty	0.7201	Thermal duty
u_{peak}	0.53728	Peak utilisation
u_{rms}	0.90012	Continuous utilisation
u_{speed}	0.9	Speed utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	RMS torque $SF \cdot T_{rms} \le T_{rated}$	u_{rms} = 0.90012	$u_{rms} \le 1$	0.9	
PASS	Speed $n_{pk} \le n_{max}$	u_{speed} = 0.9	$u_{speed} \le 1$	0.9	
PASS	Peak torque $SF \cdot T_{pk} \le T_{peak}$	u_{peak} = 0.53728	$u_{peak} \le 1$	0.537	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: J_{motor}, T_{rated}, T_{peak}, n_{max} (Candidate motor rating). Enter each as one so its source, edition and licence travel with it.
WARNING — input.T_L: T_L (Constant load torque at the output (rotary)) was not entered; the template default 0 N*m is used.
WARNING — input.duty_mode: $duty_{mode}$ (Duty (point-to-point move cycle, or continuous rotary running with a run-up)) was not entered; the template default 'point-to-point' is used.
WARNING — overhauling: T_{dec}: the motor brakes during deceleration (negative torque); the drive must absorb the regenerated energy.

8. Verification

```
Template elec.motor-sizing@1.6.0 Content hash 4ea5e25d0b648caa Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Hughes A, Drury B. Electric Motors and Drives: Fundamentals, Types and Applications. 5th ed. Newnes; 2019.
2. Kollmorgen; Parker Hannifin. Servo motor sizing guides [manufacturer technical literature].